5.3.3 Ellipse

We invoke that an ellipse is the locus of a point which moves such that its distance from a fixed point (focus) bears a constant ratio (eccentricity) less than unity its distance from its directrix bearing a constant ratio e $(0<e<1)$.

(i) Equation of an Ellipse in standard form

Let S be a focus, l be a directrix, e be the eccentricity $(0<e<1)$ and $P(x,y)$ be the moving point. Draw SZ and PM perpendicular to l.

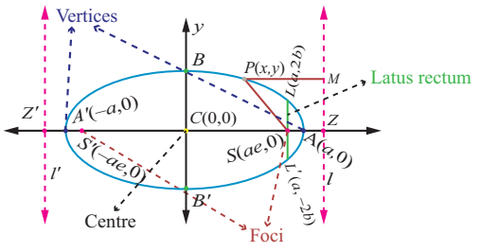


Fig.5.23

Let A and A' be the points which divide SZ internally and externally in the ratio $e:1$ respectively. Let $AA'=2a$. Let the point of intersection of the perpendicular bisector with AA' be C. Therefore $CA=a$ and $CA'=a$. Choose C as origin and CZ produced as x-axis and the perpendicular bisector of AA' produced as y-axis.

By definition,

$$\frac{SA}{AZ}=\frac{e}{1} \qquad \text{and} \qquad \frac{SA'}{A'Z}=\frac{e}{1}$$

$$SA=eAZ \qquad SA'=eA'Z$$

$$CA-CS=e(CZ-CA) \qquad A'C+CS=e(A'C+CZ)$$

$$a-CS=e(CZ-a) \quad \ldots(1) \qquad a+CS=e(a+CZ) \quad \ldots(2)$$

$(2)+(1)$ gives $CZ=\frac{a}{e}$ and $(2)-(1)$ gives $CS=ae$.

Therefore M is $\left(\frac{a}{e},y\right)$ and S is $(ae,0)$.

$$\text{By the definition of a conic, } \frac{SP}{PM}=e \Rightarrow SP^2=e^2PM^2$$

$$\Rightarrow (x-ae)^2+(y-0)=e^2\left[\left(x-\frac{a}{e}\right)^2+0\right] \text{ which}$$

on simplification yields $\frac{x^2}{a^2}+\frac{y^2}{a^2(1-e^2)}=1$.

Since $1-e^2$ is a positive quantity, write $b^2=a^2(1-e^2)$

Taking $ae=c$, $b^2=a^2-c^2$.

Hence we obtain the locus of P as $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ which is the equation of an **ellipse in standard form** and note that it is symmetrical about x and y axis.

Definition 5.4

(1) The line segment AA' is called the **major axis** of the ellipse and is of length $2a$.

(2) The line segment BB' is called the **minor axis** of the ellipse and is of length $2b$.

(3) The line segment $CA=$ the line segment $CA'=$ **semi major axis $=a$** and the line segment $CB=$ the line segment $CB'=$ **semi minor axis $=b$**.

(4) By symmetry, taking $S'(-ae,0)$ as focus and $x=-\frac{a}{e}$ as directrix l' gives the same ellipse.

Thus, we see that an ellipse has two foci, $S(ae,0)$ and $S'(-ae,0)$ and two vertices $A(a,0)$ and $A'(-a,0)$ and also two directrices, $x=\frac{a}{e}$ and $x=-\frac{a}{e}$.

Chapter 5 Analytical Geometry.indd 186 31-01-2020 17:48:03